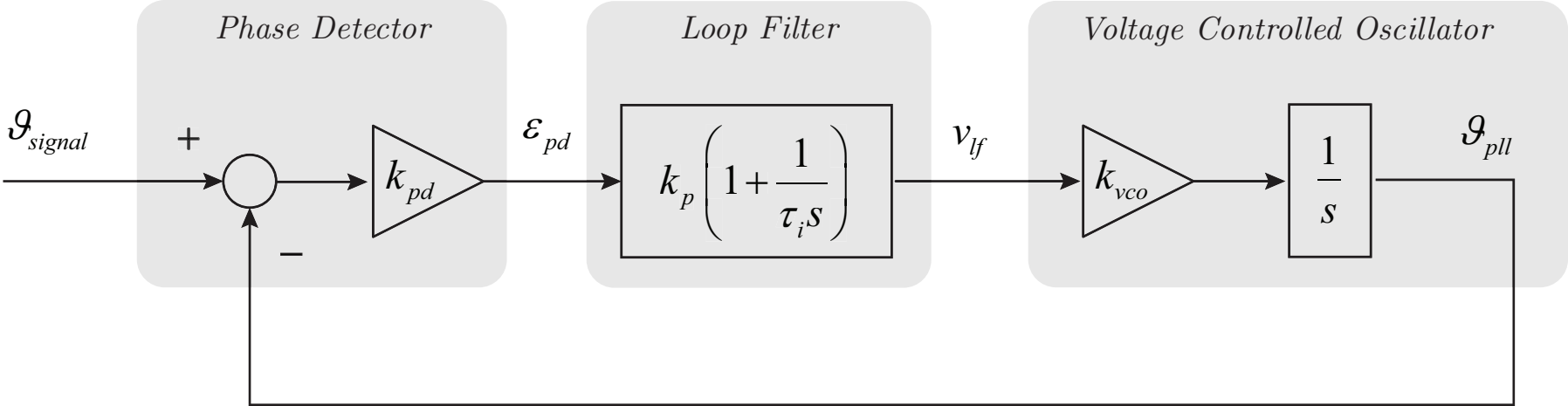
Phase Detector
Loop Filter
Voltage Controlled Oscillator
ϑ_{signal}
+
−
k_{pd}
ε_{pd}
$k_p\left(1+\frac{1}{\tau_i s}\right)$
v_{lf}
k_{vco}
$\frac{1}{s}$
ϑ_{pll}